

Annealed Grade 23 Titanium

Annealed grade 23 titanium is grade 23 titanium in the annealed condition. The graph bars on the material properties cards below compare annealed grade 23 titanium to: wrought titaniums (top), all titanium alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Elastic (Young's, Tensile) Modulus

110 GPa 15×10^6 psi

Elongation at Break

11 %

Fatigue Strength

500 MPa 72×10^3 psi

Poisson's Ratio

0.32

Reduction in Area

30 %

Rockwell C Hardness

32

Shear Modulus

40 GPa 5.8×10^6 psi

Shear Strength

570 MPa 82×10^3 psi

Tensile Strength: Ultimate (UTS)

940 MPa 140×10^3 psi

Tensile Strength: Yield (Proof)

850 MPa 120×10^3 psi

Thermal Properties

Latent Heat of Fusion

410 J/g

Maximum Temperature: Mechanical

340 °C 640 °F

Melting Completion (Liquidus)

1610 °C 2930 °F

Melting Onset (Solidus)

1560 °C 2840 °F

Specific Heat Capacity

560 J/kg-K 0.13 BTU/lb-°F

Thermal Conductivity

7.1 W/m-K 4.1 BTU/h-ft-°F

Thermal Expansion

9.4 µm/m-K

Electrical Properties

Electrical Conductivity: Equal Volume

1.0 % IACS

Electrical Conductivity: Equal Weight (Specific)

2.0 % IACS

Otherwise Unclassified Properties

Base Metal Price

36 % relative

Density

4.4 g/cm³ 280 lb/ft³

Embodied Carbon

38 kg CO₂/kg material

Embodied Energy

610 MJ/kg 260 x 10³ BTU/lb

Embodied Water

200 L/kg 23 gal/lb

Common Calculations

Resilience: Ultimate (Unit Rupture Work)

100 MJ/m³

Resilience: Unit (Modulus of Resilience)

3430 kJ/m³

Stiffness to Weight: Axial

13 points

Stiffness to Weight: Bending

35 points

Strength to Weight: Axial

59 points

Strength to Weight: Bending

48 points

Thermal Diffusivity

2.9 mm²/s

Thermal Shock Resistance

68 points